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ANATOMICALLY CONTOURED STIMULATION LEADS FOR HIGH DENSITY NEURAL INTERFACE

Abstract

The present disclosure relates to implantable neuromodulation devices and methods of fabrication, and in particular to anatomically contoured spinal cord stimulation leads for high density neural interfaces and methods of microfabricating the stimulation leads. Particularly, aspects are directed to a thin film lead assembly that includes a cable having: a first supporting structure formed of dielectric material, a first set of conductive traces formed on the first supporting structure, a second supporting structure formed of dielectric material, and a second set of conductive traces formed on the second supporting structure. The thin film lead assembly also includes an electrode assembly having: a third supporting structure formed of dielectric material, a first set of electrodes in electrical connection with the first set of conductive traces, and a second set of electrodes in electrical connection with the second set of conductive traces.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/774,397, filed May 4, 2022, which is a 371 National Phase of International Patent Application No. PCT/US2020/058713, filed Nov. 3, 2020, which claims the benefit of the filing date of U.S. Provisional Patent Application No. 62/930,457, filed Nov. 4, 2019. The disclosures of the above-named applications are incorporated by reference herein in their entireties.

FIELD OF THE INVENTION

[0002] The present disclosure relates to implantable neuromodulation devices and methods of fabrication, and in particular to anatomically contoured spinal cord stimulation leads for high density neural interfaces and methods of microfabricating the stimulation leads.

BACKGROUND

[0003] Normal neural activity is an intricate balance of electrical and chemical signals which can be disrupted by a variety of insults (genetic, chemical or physical trauma) to the nervous system, causing cognitive, motor and sensory impairments. Similar to the way a cardiac pacemaker or defibrillator corrects heartbeat abnormalities, neuromodulation therapies help to reestablish normal neural balance. In particular instances, neuromodulation therapies utilize medical device technologies to enhance or suppress activity of the nervous system for the treatment of disease. These technologies include implantable as well as non-implantable neuromodulation devices and systems that deliver electrical, chemical or other agents to reversibly modify brain and nerve cell activity. The most common neuromodulation therapy is spinal cord stimulation to treat chronic neuropathic pain. In addition to chronic pain relief, some examples of neuromodulation therapies include deep brain stimulation for essential tremor, Parkinson's disease, dystonia, epilepsy and psychiatric disorders such as depression, obsessive compulsive disorder and Tourette syndrome; sacral nerve stimulation for pelvic disorders and incontinence; vagus nerve stimulation for rheumatoid arthritis; gastric and colonic stimulation for gastrointestinal disorders such as dysmotility or obesity; vagus nerve stimulation for epilepsy, obesity or depression; carotid artery stimulation for hypertension, and spinal cord stimulation for ischemic disorders such as angina and peripheral vascular disease.

[0004] Neuromodulation devices and systems tend to have a similar form factor, derived from their predecessors, e.g. the pacemaker or defibrillator. Such neuromodulation devices and systems typically consist of an implant containing electronics connected to leads that deliver electrical pulses to electrodes interfaced with nerves or nerve bundles via an electrode assembly. The electrode assembly may be formed of a conductive material and typically take the form of book electrodes, cuff electrodes, spiral cuff electrodes, epidural electrodes, helical electrodes, probe electrodes, linear electrodes, paddle electrodes, and intraneural electrodes.

[0005] Conventional electrode assemblies include between four and thirty-two electrodes, and thus typically include four to thirty-two channels or wires connected respectively to the electrodes at the distal end and the electronics of the neurostimulator at the proximal end. However, there is a need for high density neural interfaces that include greater than thirty-two electrodes to interface with larger tissue volumes (e.g., wrapped around the dural sheath of the spinal cord or sub-dural directly

on the spinal cord), to recruit smaller populations of neurons for recording, or to provide more targeted therapy by tailoring the electrical stimulation parameters and activated tissue volume. Increasing the density or number of electrodes can increase the number of channels or wires needed to connect the electrodes and the electronics of the neurostimulator. In order to implement high channel or wire counts, there is a need for reliable electrical connections between the electrodes and leads that can maintain contact and electrical isolation in a subject body (e.g., a patient body) for many years. Conventionally, a lead assembly containing a high channel or wire count has the channels or wires connected respectively to the electrodes in a permanent manner. However, these lead assemblies are typically limited in design complexity including channel count and anatomical compliance. Therefore, there is a need for reliable connectors for lead assemblies having high density and anatomically compliant neural interfaces.

BRIEF SUMMARY

[0006] In various embodiments, a thin-film lead assembly is provided including: a cable including: (i) a first supporting structure formed of one or more layers of dielectric material; (ii) a first set of conductive traces formed on a portion of the first supporting structure; (iii) a second supporting structure formed of one or more layers of dielectric material; and (iv) a second set of conductive traces formed on a portion of the second supporting structure; and an electrode assembly including: (i) a third supporting structure formed of one or more layers of dielectric material, (ii) a first set of electrodes in electrical connection with the first set of conductive traces, and (iii) a second set of electrodes in electrical connection with the second set of conductive traces.

[0007] In some embodiments, the first set of electrodes and the second set of electrodes are arranged on the third supporting structure in a honeycomb pattern.

[0008] In some embodiments, the first set of electrodes and the second set of electrodes are arranged on the third supporting structure in a plurality of alternating rows including a first number of electrodes n and a second number of electrodes $n-1$.

[0009] In some embodiments, the electrode assembly further includes a proximal end and a distal end, the first supporting structure and the second supporting structure connect with the electrode assembly at the proximal end, and the row of electrodes closest to the first supporting structure and the second supporting structure at the proximal end includes a third number of electrodes $n-2$.

[0010] In some embodiments, the third supporting structure is thermoset into a cylindrical shape.

[0011] In some embodiments, the first set of conductive traces and the second set of conductive traces include one or more layers of conductive material, and the conductive material is copper (Cu), gold (Au), silver (Ag), gold/chromium (Au/Cr), platinum (Pt), platinum/iridium (Pt/Ir), titanium (Ti), gold/titanium (Au/Ti), or any alloy thereof.

[0012] In some embodiments, the one or more layers of dielectric material of the first supporting structure, the second supporting structure, and the third supporting structure are monolithic, and the dielectric material is polyimide, liquid crystal polymer, parylene, polyether ether ketone, or a combination thereof.

[0013] In some embodiments, the first supporting structure includes a proximal end and a distal end, the second supporting structure includes a proximal end and a distal end, the electrode assembly further includes a proximal end and a distal end, the distal end of the first supporting structure and the distal end of the second supporting structure are continuous with the proximal end of electrode assembly.

[0014] In some embodiments, the thin-film lead assembly further includes: a first connector formed on the proximal end of the first supporting structure, where the first connector includes a first set of contacts or bond pads in electrical connection with the first set of conductive traces; and a second connector formed on the proximal end of the second supporting structure, where the second connector includes a second set of contacts or bond pads in electrical connection with the second set of conductive traces.

[0015] In some embodiments, the first connector is a branched connector including: a main body

including the first supporting structure and the first set of conductive traces; and a plurality of plugs extending from the main body, each plug of the plurality of plugs includes: an end portion of the first supporting structure includes the one or more layers of dielectric material; and a subset of conductive traces from the first set of conductive traces, where each trace from the subset of conductive traces terminates at a contact or bond pad of the first set of contacts or bond pads exposed on a surface of the end portion of the first supporting structure.

[0016] In some embodiments, the second connector is a branched connector including: a main body including the second supporting structure and the second set of conductive traces; and a plurality of plugs extending from the main body, each plug of the plurality of plugs includes: an end portion of the second supporting structure includes the one or more layers of dielectric material; and a subset of conductive traces from the second set of conductive traces, where each trace from the subset of conductive traces terminates at a contact or bond pad of the second set of contacts or bond pads exposed on a surface of the end portion of the second supporting structure.

[0017] In some embodiments, the first supporting structure include a first helical portion and the second supporting structure includes a second helical portion, the first supporting structure and the second supporting structure run in parallel adjacent to one another within the cable.

[0018] In some embodiments, the cable further includes a housing encasing the helical portion of the first supporting structure and the helical portion of the second supporting structure.

[0019] In various embodiments, a thin-film lead assembly is provided including: a first cable including: (i) a first supporting structure formed of one or more layers of dielectric material, and (ii) a first set of conductive traces formed on a portion of the first supporting structure; a second cable including: (i) a second supporting structure formed of one or more layers of the dielectric material, and (ii) a second set of conductive traces formed on a portion of the second supporting structure; a first electrode assembly including: (i) a backing formed of a polymer material, (ii) a third supporting structure formed of one or more layers of dielectric material, (iii) a first set of electrodes in electrical connection with first set of conductive traces; and a second electrode assembly including: (i) the backing formed of the polymer material, (ii) a fourth supporting structure formed of one or more layers of dielectric material, and (iii) a second set of electrodes in electrical connection with the second set of conductive traces.

[0020] In some embodiments, the third supporting structure and the fourth supporting structure are formed on the backing, the backing is formed of thermoplastic polyurethane, and the backing, the third supporting structure, and the fourth supporting structure are thermoset into a cylindrical shape.

[0021] In some embodiments, the one or more layers of dielectric material of the first supporting structure and the third supporting structure are monolithic, the one or more layers of dielectric material of the second supporting structure and the fourth supporting structure are monolithic, and the dielectric material is polyimide, liquid crystal polymer, parylene, polyether ether ketone, or a combination thereof.

[0022] In some embodiments, the first electrode assembly further includes (iv) a first set of alignment holes formed in the third supporting structure, the second electrode assembly further includes (iv) a second set of alignment holes formed in the fourth supporting structure, and the first set of alignment holes and the second set of alignment holes are aligned with one another.

[0023] In some embodiments, the first supporting structure further includes a proximal end and a distal end, the second supporting structure further includes a proximal end and a distal end, the first electrode assembly further includes a proximal end and a distal end, the second electrode assembly further includes a proximal end and a distal end, the distal end of the first supporting structure is continuous with the proximal end of the first electrode assembly, and the distal end of the second supporting structure is continuous with the proximal end of the second electrode assembly.

[0024] In some embodiments, the thin-film lead assembly further includes: a first connector formed on the proximal end of the first supporting structure, where the first connector includes a first set of

contacts or bond pads in electrical connection with the first set of conductive traces; and a second connector formed on the proximal end of the second supporting structure, where the second connector includes a second set of contacts or bond pads in electrical connection with the second set of conductive traces.

[0025] In various embodiments, a neuromodulation system is provided including: a neurostimulator including an electronics module; a cable including: (i) a first supporting structure formed of one or more layers of dielectric material; (ii) a first set of conductive traces formed on a portion of the first supporting structure; (iii) a second supporting structure formed of one or more layers of dielectric material; and (iv) a second set of conductive traces formed on a portion of the second supporting structure; an electrode assembly including: (i) a third supporting structure formed of one or more layers of dielectric material, (ii) a first set of electrodes in electrical connection with the first set of conductive traces, and (iii) a second set of electrodes in electrical connection with the second set of conductive traces; a first connector formed on a proximal end of the first supporting structure, where the first connector includes a first set of contacts or bond pads in electrical connection with the first set of conductive traces and the electronics module; and a second connector formed on a proximal end of the second supporting structure, where the second connector includes a second set of contacts or bond pads in electrical connection with the second set of conductive traces and the electronics module.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The present invention will be better understood in view of the following non-limiting figures, in which:

[0027] FIG. 1 shows a neuromodulation system in accordance with various embodiments;

[0028] FIG. 2 shows an electrode assembly in accordance with various embodiments;

[0029] FIGS. 3A, 3B, and 3C show an anatomically contoured electrode assembly in accordance with various embodiments;

[0030] FIGS. 4A and 4B show a lead assembly in accordance with various embodiments;

[0031] FIGS. 5A, 5B, 5C, and 5D show an alternative electrode assembly in accordance with some aspects of the present invention; and

[0032] FIG. 6 shows electrode assemblies in accordance with some aspects of the present invention.

DETAILED DESCRIPTION

I. Introduction

[0033] The following disclosure describes anatomically contoured spinal cord stimulation leads for high density neural interfaces and methods of microfabricating the stimulation leads. As used herein, the term “proximal” or “proximal end” refers to a first end of the main body, while the term “distal” or “distal end” refers to a second end opposing the first end. For example, the proximal end may be an end of the main body, which is closest to the user, and the distal end may be an end of the main body, which is furthest from the user. The stimulation leads may be fabricated using microfabricating techniques. In certain embodiments, the stimulation leads are fabricated as a monolithic structure. As used herein, the phrase “monolithic” refers to a device fabricated using a same layer of base material. As used herein, the phrase “microfabrication” refers to the process of fabricating miniature structures on micrometer scales and smaller. The major concepts and principles of microfabrication are microlithography, doping, thin films, etching, bonding, and polishing. As used herein, the phrase “thin films” refers to a layer of material ranging from fractions of a nanometer (monolayer) to several micrometers in thickness (e.g., between a few nanometers to about 100 μm , or the thickness of a few atoms). Thin films may be deposited by

applying a very thin film of material (e.g., between a few nanometers to about 100 μm , or the thickness of a few atoms) onto a substrate surface to be coated, or onto a previously deposited layer of thin film. In various embodiments, a thin film lead assembly is provided including a base polymer body (e.g., a supporting structure) and at least one conductive trace formed on the base polymer body. As used herein, the term “high density neural interface(s)” refers to a neural interface that includes at least thirty-two electrodes (i.e., recording, sensing, stimulating, other types of electrodes, or combinations thereof).

[0034] Neuromodulation devices such spinal cord stimulators electrically interface with neural tissue and treat various neurological conditions through electrical stimulation. As described herein, conventional neuromodulation devices include a neurostimulator and lead assembly containing between four and thirty-two electrodes. There is a need for high-density lead assemblies that can significantly increase the number of electrodes in order to interface with larger tissue volume, to recruit smaller populations of neurons for recording, or to provide more targeted therapy by tailoring the electrical stimulation parameters and activated tissue volume. Conventional neuromodulation is limited to no more than thirty-two channels per lead assembly due partly to a lack of compatible lead and electrode assembly technology.

[0035] To address these limitations and problems, lead assemblies of various embodiments disclosed herein include high density neural interfaces that are capable of being anatomically contoured to a biological system such as a spinal cord. One illustrative embodiment of the present disclosure is directed to a thin film lead assembly including: a cable including: (i) a first supporting structure formed of one or more layers of dielectric material; (ii) a first set of conductive traces formed on a portion of the first supporting structure; (iii) a second supporting structure formed of one or more layers of dielectric material; and (iv) a second set of conductive traces formed on a portion of the second supporting structure; and an electrode assembly including: (i) a third supporting structure formed of one or more layers of dielectric material, (ii) a first set of electrodes in electrical connection with the first set of conductive traces, and (iii) a second set of electrodes in electrical connection with the second set of conductive traces.

[0036] In other embodiments, a thin film lead assembly is provided that includes: a first cable including: (i) a first supporting structure formed of one or more layers of dielectric material, and (ii) a first set of conductive traces formed on a portion of the first supporting structure; a second cable including: (i) a second supporting structure formed of one or more layers of the dielectric material, and (ii) a second set of conductive traces formed on a portion of the second supporting structure; a first electrode assembly including: (i) a backing formed of a polymer material, (ii) a third supporting structure formed of one or more layers of dielectric material, (iii) a first set of electrodes in electrical connection with first set of conductive traces; and a second electrode assembly including: (i) the backing formed of the polymer material, (ii) a fourth supporting structure formed of one or more layers of dielectric material, and (iii) a second set of electrodes in electrical connection with the second set of conductive traces.

[0037] In other embodiments, neuromodulation system including: a neurostimulator including an electronics module; a cable including: (i) a first supporting structure formed of one or more layers of dielectric material; (ii) a first set of conductive traces formed on a portion of the first supporting structure; (iii) a second supporting structure formed of one or more layers of dielectric material; and (iv) a second set of conductive traces formed on a portion of the second supporting structure; an electrode assembly including: (i) a third supporting structure formed of one or more layers of dielectric material, (ii) a first set of electrodes in electrical connection with the first set of conductive traces, and (iii) a second set of electrodes in electrical connection with the second set of conductive traces; a first connector formed on a proximal end of the first supporting structure, where the first connector includes a first set of contacts or bond pads in electrical connection with the first set of conductive traces and the electronics module; and a second connector formed on a proximal end of the second supporting structure, where the second connector includes a second set

of contacts or bond pads in electrical connection with the second set of conductive traces and the electronics module.

[0038] Advantageously, these approaches provide a lead assembly, which has increased electrode count (high density neural interface), a smaller footprint, and greater design flexibility. More specifically, these approaches enable lead assemblies with reliable connections between a lead/cable and high density neural interface capable of countering to anatomical biological systems such as the spinal cord. This solution is scalable to connecting many electrodes (e.g., greater than thirty-two) using a single support system or multiple-support systems reflowed on a single backing, and thus enabling several therapeutic opportunities for neurostimulation. Furthermore even for applications where high density neural interfaces are not required, various embodiments can be miniaturized to make the implant minimally invasive, additionally may make invasive anatomies to become accessible (or navigable) due to the miniaturization. It should be understood that although spinal cord neurostimulation and spinal cord device applications are provided as examples of some embodiments, this solution is applicable to all leads and devices that need electrodes/sensors that need to be attached to a neurostimulator.

II. Neuromodulation Devices and Systems with an Electrode Assembly

[0039] FIG. 1 shows a neuromodulation system **100** in accordance with some aspects of the present invention. In various embodiments, the neuromodulation system **100** includes an implantable neurostimulator **105** and a lead assembly **110**. The implantable neurostimulator **105** may include a housing **115**, a feedthrough assembly **120**, a power source **125**, an antenna **130**, and an electronics module **135** (e.g., a computing system). The housing **115** may include materials that are biocompatible such as bioceramics or bioglasses for radio frequency transparency, or metals such as titanium. In accordance with some aspects of the present invention, the size and shape of the housing **115** may be selected such that the neurostimulator **105** can be implanted within a patient. In the example shown in FIG. 1, the feedthrough assembly **120** is attached to a hole in a surface of the housing **115** such that the housing **115** is hermetically sealed. The feedthrough assembly **120** may include one or more feedthroughs (i.e., electrically conductive elements, pins, wires, tabs, pads, etc.) mounted within and extending through the surface of the housing **115** or a cap from an interior to an exterior of the housing **115**. The power source **125** may be within the housing **115** and connected (e.g., electrically connected) to the electronics module **135** to power and operate the components of the electronics module **135**. The antenna **130** may be connected (e.g., electrically connected) to the electronics module **135** for wireless communication with external devices via, for example, radiofrequency (RF) telemetry.

[0040] In some embodiments, the electronics module **135** may be connected (e.g., electrically connected) to interior ends of the feedthrough assembly **120** such that the electronics module **135** is able to apply a signal or electrical current to leads of the lead assembly **110** connected to exterior ends of the feedthrough assembly **120**. The electronics module **135** may include discrete and/or integrated electronic circuit components that implement analog and/or digital circuits capable of producing the functions attributed to the neuromodulation devices or systems such as applying or delivering neural stimulation to a patient. In various embodiments, the electronics module **135** may include software and/or electronic circuit components such as a pulse generator **140** that generates a signal to deliver a voltage, current, optical, or ultrasonic stimulation to a nerve or artery/nerve plexus via electrodes, a controller **145** that determines or senses electrical activity and physiological responses via the electrodes and sensors, controls stimulation parameters of the pulse generator **140** (e.g., control stimulation parameters based on feedback from the physiological responses), and/or causes delivery of the stimulation via the pulse generator **140** and electrodes, and a memory **150** with program instructions operable on by the pulse generator **140** and the controller **145** to perform one or more processes for applying or delivering neural stimulation.

[0041] The lead assembly **110** may include a lead body **155**, a lead connector **160**, an electrode assembly **165**, and optionally one or more sensors. In some embodiments, the lead connector **160**

may be bonding material that bonds conductor material of the lead body **155** to the electronics module **135** of the implantable neurostimulator **105** via the feedthrough assembly **120**. The bonding material may be a conductive epoxy or a metallic solder or weld such as platinum. In other embodiments, the lead connector **160** may be conductive wire or tab (e.g., a wire or tab formed of copper, silver, or gold) that bonds conductor material of the lead body **155** to the electronics module **135** of the implantable neurostimulator **105**. In alternative embodiments, the implantable neurostimulator **105** and the lead body **155** may be designed to connect with one another via a lead connector **160** such as a branched connector, pin and sleeve connector, snap and lock connector, flexible printed circuit connector, or combination thereof.

[0042] The lead body **155** may include one or more leads **170** of conductive material and dielectric material. The one or more leads **170** may carry electrical conductors that allow electrical coupling of the electronics module **135** to electrodes **180** of the electrode assembly **165** via the lead connector **160**. In some examples the one or more leads **170** are formed of one or more layers of dielectric material such as a polymer having suitable dielectric, flexibility and biocompatibility characteristics. Polyurethane, polycarbonate, silicone, polyethylene, fluoropolymer and/or other medical polymers, copolymers and combinations or blends can be used. In some embodiments, the dielectric material is polyimide, liquid crystal polymer, parylene, polyether ether ketone, or a combination thereof. The conductive material may be any suitable conductor such as stainless steel, silver, copper or other conductive materials, which may have separate coatings or sheathing for anticorrosive, insulative and/or protective reasons. In certain embodiments, the conductive material copper (Cu), gold (Au), silver (Ag), gold/chromium (Au/Cr), platinum (Pt), platinum/iridium (Pt/Ir), titanium (Ti), gold/titanium (Au/Ti), or any alloy thereof. The conductive material may take various forms including wires, drawn filled tubes, helical coiled conductors, microwires, and/or printed circuits (i.e., conductive traces), for example.

[0043] FIG. 2 shows an electrode assembly **200** in accordance with some aspects of the present invention. In various embodiments, the electrode assembly **200** may include a plurality of electrodes **205** and an supporting structure **210** fabricated in accordance with various aspects. The electrode assembly **200** may be connected to conductor material of a lead body or cable **215** (e.g., a lead body **155** as described with respect to FIG. 1) via one or more leads (e.g., one or more leads **170** as described with respect to FIG. 1). In some embodiments, an end of the cable **215** carries the electrode assembly **200** (e.g., book electrodes, split cuff electrodes, spiral cuff electrodes, epidural electrodes, helical electrodes, probe electrodes, linear electrodes, neural probe, paddle electrodes, and intraneural electrodes). In other embodiments, an end of the cable **215** carries a plurality of electrode assemblies **200**.

[0044] The supporting structure **210** may include one or more layers of dielectric material, for example, polyimide, liquid crystal polymer, parylene, polyether ether ketone, or a combination thereof. The one or more layers of dielectric material provide support for microelectronic structures including the plurality of electrodes **205** and a wiring layer (embedded within the one or more dielectric layers). The plurality of electrodes **205** may be electrically connected to the wiring layer via one or more via or contacts, and the wiring layer is used to electrically connect the plurality of electrodes **205** to the one or more leads of the cable **215**. The plurality of electrodes **205** and wiring layer may be formed of one or more layers of conductive material. In some instances, the conductive material is copper (Cu), gold (Au), silver (Ag), titanium (Ti), or platinum (Pt), or alloys thereof such as gold/chromium (Au/Cr) or Titanium/Platinum (Ti/Pt), for example. In some embodiments, the plurality of electrodes **205** include a top surface that is raised above the top surface of the supporting structure **210** by a predetermined distance. The predetermined distance may be greater than 0.1 μm , greater than 0.5 μm , greater than 1 μm or greater than 10 μm , or the predetermined distance may be from 0.1 μm to 50 μm , from 0.3 μm to 40 μm , from 0.5 μm to 30 μm , or from 1 μm to 25 μm . In other embodiments, the plurality of electrodes **205** include a top surface that is coplanar with the top surface of the supporting structure **210**. The plurality of

electrodes **205** may be formed directly on the supporting structure **210**. The term “directly”, as used herein, may be defined as being without something in between. Alternatively, the plurality of electrodes **205** may be formed indirectly on the supporting structure **210**. The term “indirectly”, as used herein, may be defined as having something in between.

[0045] The cable **215** may include: (i) one or more supporting structures **220** (e.g., one or more leads) formed of one or more layers of dielectric material, for example, polyimide, liquid crystal polymer, parylene, polyether ether ketone, or a combination thereof, and (ii) one or more sets of conductive traces **225** formed on the one or more supporting structures **220** (e.g., one or more leads), respectively. The sets of conductive traces **225** may include one or more layers of conductive material, and the conductive material may be copper (Cu), gold (Au), silver (Ag), gold/chromium (Au/Cr), platinum (Pt), platinum/iridium (Pt/Ir), titanium (Ti), gold/titanium (Au/Ti), or any alloy thereof. In some instances, the one or more layers of dielectric material of the supporting structure **210** and the one or more layers of dielectric material of the one or more supporting structures **220** are monolithic. More specifically, each of the one or more supporting structures **220** includes a proximal end **230** and a distal end **235**, the electrode assembly **200** further includes a proximal end **240** and a distal end **245**, and the distal end **235** of the one or more supporting structures **220** may be continuous with the proximal end **240** of the electrode assembly **200**.

[0046] The plurality of electrodes **205** may be separated into two or more sets, e.g., a first set of electrodes (A) and a second set of electrodes (B). The first set of electrodes (A) may be formed on the supporting structure **210** in electrical connection with a first set of conductive traces (X) of the cable **215**. The second set of electrodes (B) may be formed on the supporting structure **210** in electrical connection with a second set of conductive traces (Y) of the cable **215**. The two or more sets of electrodes (e.g., (A) and (B)) may be arranged on the supporting structure **210** in a predetermined pattern. In some instances, the predetermined pattern is a honeycomb pattern where the first set of electrodes (A) and the second set of electrodes (B) are arranged on the supporting structure **210** in a plurality of alternating rows including a first number of electrodes n (e.g., 6) and a second number of electrodes $n-1$ (e.g., 5). The distal end **235** of the cable **215** connects or is integrated with the proximal end **240** of the electrode assembly **200**. In certain instances, a row of electrodes (C) closest to the cable **215** at the proximal end **240** includes a third number of electrodes $n-2$ (e.g., 4) to provide more routing flexibility for the wiring layers and conductive traces.

[0047] In various embodiments, the supporting structure **210** has a horizontal width (W) and vertical height (H). The horizontal width (W) may range from 5 mm to 50 mm, from 10 mm to 30 mm, from 5 mm to 25 mm, or from 15 mm to 25 mm (for example, 19 mm). The vertical height (h) may range from 5 mm to 90 mm, from 20 mm to 60 mm, from 20 mm to 45 mm, or from 30 mm to 50 mm (for example, 33 mm). Retaining the horizontal width (W) while spacing out the numbers of electrodes n , $n-1$, and $n-2$ an equal distance, the supporting structure **210** may be configured to provide a center-to-center spacing (C) for the plurality electrodes **205** that ranges from 0.5 mm to 10 mm, from 0.5 mm to 5 mm, from 1 mm to 8 mm, or from 1 mm to 5 mm (for example, 2.9 mm). The plurality of electrodes **205** may be formed in a predetermined shape on the supporting structure **210**. In some instances, the predetermined shape of the plurality of electrodes **205** is a circle having a diameter (D) that ranges from 0.5 mm to 5 mm, from 0.5 mm to 4 mm, from 1 mm to 4 mm, or from 1 mm to 3 mm (for example, 1.9 mm). The plurality of electrodes **205** may have a thickness (T) that ranges from 0.1 μm to 50 μm , from 0.3 μm to 30 μm , from 0.5 μm to 20 μm , or from 1 μm to 15 μm . The plurality of rows of plurality of electrodes **205** may have a spacing (K) that ranges from 0.5 mm to 5 mm, from 0.5 mm to 4 mm, from 1 mm to 4 mm, or from 1 mm to 3 mm (for example, 1.0 mm).

[0048] FIGS. 3A-3C show electrode assembly **300** (e.g., the electrode assembly as described with respect to FIG. 2) in accordance with various aspects. In various embodiments, the electrode

assembly **300** may include one or more electrodes **305** and a supporting structure **310** fabricated in accordance with various aspects. The electrode assembly **300** may be connected to conductor material of a lead body or cable **315** (e.g., a lead body **155** as described with respect to FIG. **1**) via one or more leads (e.g., one or more leads **170** as described with respect to FIG. **1**). In some embodiments, the supporting structure **310** may include one or more layers of dielectric material formed in the shape of a structure (e.g., an arc, curl, cylinder, etc.) that allows for the electrode assembly **300** to be opened or closed in order to position the electrode assembly **300** on a nerve or artery/nerve plexus and to allow for the electrode assembly **300** to move with tissue of a patient's body. The one or more layers of dielectric material may also provide support for microelectronic structures including the one or more electrodes **305**, wiring layers, and optional vias or contacts. In certain instances, the one or more layers of dielectric material are thermoset with a curled portion **320** having a predetermined radius (R). The radius (R) may be predetermined based on the anatomical structure intended to interface with the electrode assembly **300**. For example, the curled portion **320** may have a radius (R) that ranges from 3.5 mm to 7 mm or from 4 mm to 6.5 mm (e.g., 6 mm) to conform with the average human computed tomography spinal cord measurements. In some instances, the radius (R) may be predetermined to fit around the dural sheath between the vertebrae, or could be used sub-dural (beneath the dura, in direct contact with the spinal cord). [0049] FIG. **4A** shows a lead assembly **400** (e.g., the monolithic lead assembly **110** described with respect to FIG. **1**) in accordance with aspects of the present disclosure. In various embodiments, the lead assembly **400** includes one or more cables **405** having a proximal end **410** and a distal end **415**. Each cable **405** may include one or more supporting structures **420** and one or more conductive traces **425** formed on a portion of the one or more supporting structures **420**. As used herein, the term “formed on” refers to a structure or feature that is formed on a surface of another structure or feature, a structure or feature that is formed within another structure or feature, or a structure or feature that is formed both on and within another structure or feature. In some embodiments, the supporting structure **420** extends from the proximal end **410** to the distal end **415**. In some embodiments, the supporting structure **420** may be made of one or more layers of dielectric material (i.e., an insulator). The dielectric material may be selected from the group of electrically nonconductive materials consisting of organic or inorganic polymers, ceramics, glass, glass-ceramics, polyimide-epoxy, epoxy-fiberglass, and the like. In certain embodiments, the dielectric material is a polymer of imide monomers (i.e., a polyimide), a liquid crystal polymer (LCP) such as Kevlar®, parylene, polyether ether ketone (PEEK), or combinations thereof. In other embodiments, the supporting structure **420** may be made of one or more layers of dielectric material formed on a substrate. The substrate may be made from any type of metallic or non-metallic material.

[0050] In various embodiments, the supporting structure **420** including the one or more layers of dielectric material, and optionally the substrate, has a thickness (G) from the proximal end **410** to the distal end **415**. In some embodiments, the thickness (G) is from 10 μm to 150 μm , for example about 50 μm or about 60 μm . As used herein, the terms “substantially,” “approximately” and “about” are defined as being largely but not necessarily wholly what is specified (and include wholly what is specified) as understood by one of ordinary skill in the art. In any disclosed embodiment, the term “substantially,” “approximately,” or “about” may be substituted with “within [a percentage] of” what is specified, where the percentage includes 0.1, 1, 5, and 10 percent. In some embodiments, the supporting structure **420** has a length (L) of 5 cm to 150 cm or 50 cm to 100 cm, e.g., about 75 cm. In some embodiments, the supporting structure **420** has a width (Z) from a first side horizontally to a second side. In some embodiments, the width (Z) is from 25 μm to 5 mm, for example about 400 μm or about 1000 μm .

[0051] In various embodiments, the one or more conductive traces **425** are a plurality of traces, for example, two or more conductive traces or from two to thirty-two conductive traces. The plurality of conductive traces **425** include one or more layers of conductive material. The conductive

material selected for the one or more conductive traces **425** should have good electrical conductivity and may include pure metals, metal alloys, combinations of metals and dielectrics, and the like. For example, the conductive material may be copper (Cu), gold (Au), silver (Ag), gold/chromium (Au/Cr), etc. In some embodiments, it is also desirable that the conductive material selected for the one or more conductive traces **425** have thermal expansion characteristics or a coefficient of thermal expansion (CTE) that is approximately equal to that of CTE of the supporting structure **420**. Matching the CTE of components that contact one another is desirable because it eliminates the development of thermal stresses, which may occur during fabrication and the operation of the cable, and thus eliminates a known cause of mechanical failure in the components.

[0052] The one or more conductive traces **425** may be deposited onto a surface of the supporting structure **420** by using thin film deposition techniques well known to those skilled in the art such as by sputter deposition, chemical vapor deposition, metal organic chemical vapor deposition, electroplating, electroless plating, and the like. In some embodiments, the thickness of the one or more conductive traces **425** is dependent on the particular impedance desired for conductor, in order to ensure excellent signal integrity (e.g., electrical signal integrity for stimulation or recording). For example, if a conductor having a relatively high impedance is desired, a small thickness of conductive material should be deposited onto the supporting structure **420**. If, however, a signal plane having a relatively low impedance is desired, a greater thickness of electrically conductive material should be deposited onto the supporting structure **420**. In certain embodiments, each of the one or more conductive traces **425** has a thickness (J). In some embodiments, the thickness (J) is from 0.5 μm to 100 μm or from 25 μm to 50 μm , for example about 25 μm or about 40 μm . In some embodiments, each of the one or more conductive traces **425** has a length (M) of about 5 cm to 200 cm or 50 cm to 150 cm, e.g., about 80 cm. In certain embodiments, each of the one or more conductive traces **425** extends from the proximal end **410** to the distal end **415**. In some embodiments, each of the one or more conductive traces **425** has a width (N) from 2.0 μm to 500 μm , for example about 30 μm or about 50 μm .

[0053] As shown in FIG. **4B**, the lead assembly **400** may be formed with a predetermined shape **430** in accordance with aspects of the present disclosure. In particular, the lead assembly **400** may be formed with a predetermined shape **430** from a prefabricated wafer or panel of dielectric material or optionally a substrate. For example, the lead assembly **400** may be laser cut from a prefabricated wafer or panel in one or more spiral shapes **430**. The spiral shapes **430** may include characteristics designed to promote a monolithic fabrication process with an electrode assembly **435** and maximize the length of the lead assembly **400** that can be fabricated from a single wafer or panel. Conventionally, wafers or panels have a diameter, length, and/or width of less than 10 cm. In some embodiments, the characteristics of the spiral shapes **430** include a first spiral (A) extending from the electrode assembly **435** at a 40° to 50° angle (Q) (e.g., a 45° angle) and wrapped in a clockwise direction and second spiral (B) extending from the electrode assembly **435** at a 40° to 50° angle (O) (e.g., a 45° angle) and wrapped in a counterclockwise direction to promote a monolithic fabrication process. In some embodiments, the characteristics of the spiral shapes **430** further include a predetermined number of turns **440** and a predetermined pitch (P) between each of the turns **440** to maximize the overall length obtainable for the lead assembly **400**. In certain embodiments, the spiral shapes **430** have one or more turns, for example from 2 to 25 turns, and a pitch (P) between each of the turns from 10 μm to 1 cm or from 250 μm to 2 mm, for example about 350 μm . Accordingly, the spiral shapes can maximize the length of the lead assembly **400** that can be fabricated from a single wafer or panel. For example, a single wafer or panel with a limited diameter, length, and/or width of less than 10 cm, can be used to fabricate a lead assembly **400** with a length of 5 cm to 150 cm, 10 cm to 100 cm, or 25 cm to 75 cm, e.g., about 15 cm, using the spiral shapes **430**.

[0054] With respect back to FIG. **4A**, the lead assembly **400** further includes the electrode

assembly **435** including a supporting structure **445** that provides support for microelectronic structures including one or more electrodes **450**, a wiring layer **455**, and optional contact(s) (e.g., the electrode assemblies **200** and **300** as described with respect to FIGS. **2**, **3A**, **3B**, and **3C**). The electrode assembly **435** may be located at the distal end **415** of the lead assembly **400**. In various embodiments, the supporting structure **445** of the lead assembly and the supporting structure **420** of the one or more cables **405** are the same structure (i.e., the supporting structure is continuous from the proximal end **410** to the distal end **415**), which thus creates a monolithic lead assembly **400**. In some embodiments, the supporting structure **445** for the electrode assembly **435** includes the one or more layers of dielectric material, and optionally the substrate, and has a thickness (V) of from 10 μm to 150 μm , from 15 μm to 70 μm , from 30 μm to 60 μm , or from 40 μm to 60 μm .

[0055] The wiring layer **455** may be formed on the supporting structure **445**. In various embodiments, the wiring layer **455** is formed continuously of the one or more conductive traces **425**, and includes various metals or alloys thereof, for example, copper (Cu), gold (Au), silver (Ag), gold/chromium (Au/Cr), platinum (Pt), platinum/iridium (Pt/Ir), titanium (Ti), gold/titanium (Au/Ti), or any alloy thereof. The wiring layer **455** may have a thickness (E) of from 0.5 μm to 100 μm , from 0.5 μm to 15 μm , from 0.5 μm to 10 μm , or from 0.5 μm to 5 μm . In some embodiments, a top surface of the wiring layer **455** is coplanar with a top surface of the supporting structure **445**. In other embodiments, the wiring layer **455** is embedded within the supporting structure **445**. In yet other embodiments, the wiring layer **455** is formed on the top surface of the supporting structure **445** and the top surface of the wiring layer **455** is raised above the top surface of the supporting structure **445**.

[0056] In some embodiments, the one or more electrodes **450** are formed on the supporting structure **445** and in electrical contact with the wiring layer **455**. The one or more electrodes **450** may include conductive material such as copper (Cu), gold (Au), silver (Ag), gold/chromium (Au/Cr), platinum (Pt), platinum/iridium (Pt/Ir), titanium (Ti), gold/titanium (Au/Ti), or any alloy thereof, for example. The one or more electrodes **450** may be formed directly on the supporting structure **445**. Alternatively, the one or more electrodes **450** may be formed indirectly on the supporting structure **445**. In some embodiments, the contact(s) are formed on the supporting structure **445** and provide electrical contact between the one or more electrodes **450** and the wiring layer **455**. The contact(s) may include conductive material such as copper (Cu), gold (Au), silver (Ag), gold/chromium (Au/Cr), platinum (Pt), platinum/iridium (Pt/Ir), titanium (Ti), gold/titanium (Au/Ti), or any alloy thereof, for example.

[0057] The properties of the supporting structures and electronic structures (e.g., thickness, material, position, contact, etc.) of the lead assembly **400** may be the same or different from those of the structures previously discussed herein with reference to FIGS. **2**, **3A**, **3B**, and **3C**. However, it should be understood the lead assembly **400** is an exemplary embodiment, and that the lead assembly **400** is to be construed with the broadest sense to include variations of the specific design and/or performance need described herein, as well as other variations that are well known to those of skill in the art. In particular, the shape and location of components and layers in the lead assembly **400** may be adjusted or modified to meet specific design and/or performance needs. Furthermore, it is to be understood that other structures may have been omitted from the description of the lead assembly **400** for clarity. The omitted structures may include, for example, sensor structures, insulating layers, interconnect components, passive devices, etc.

[0058] FIG. **5A** shows an alternative lead assembly **500** (e.g., the monolithic lead assembly **110** described with respect to FIG. **1**) in accordance with aspects of the present disclosure. In various embodiments, the supporting structures and electronic structures described with respect to FIGS. **2**, **3A**, **3B**, **3C**, **4A**, and **4B** can be arranged into separate subpanels that are then stacked together and stitched into a single structure using a backing. For example, a first subpanel **505** may be provided that includes a first cable **510** including: (i) a first supporting structure **515** formed of one or more layers of dielectric material, and (ii) a first set of conductive traces **520** formed on a portion of the

first supporting structure **515**. A second subpanel **525** may be provided that includes a second cable **530** including: (i) a second supporting structure **535** formed of one or more layers of dielectric material, and (ii) a second set of conductive traces **540** formed on a portion of the second supporting structure **535**. The first subpanel **505** may further include a first electrode assembly **545** including: (i) a third supporting structure **550** formed of one or more layers of dielectric material, and (ii) a first set of electrodes **555** in electrical connection with first set of conductive traces **520**. The second subpanel **525** may further include a second electrode assembly **560** including: (i) a fourth supporting structure **565** formed of one or more layers of dielectric material, and (ii) a second set of electrodes **570** in electrical connection with second set of conductive traces **540**. [0059] In various embodiments, the third supporting structure **550** and the fourth supporting structure **565** are formed on a backing **575** to fabricate the final structure of the alternative lead assembly **500**. In some instance, the backing is formed of thermoplastic polyurethane, and the backing, the third supporting structure, and the fourth supporting structure are thermoset into a cylindrical shape, as shown in FIGS. 5B, 5C, and 5D. In some embodiments, the first electrode assembly **545** further includes (iv) a first set of alignment holes **580** formed in the third supporting structure **550**, the second electrode assembly **560** further includes (iv) a second set of alignment holes **585** formed in the fourth supporting structure **565**, and the first set of alignment holes **580** and the second set of alignment holes **585** are aligned with one another to assist in fabrication and attachment o backing **575**.

[0060] The one or more layers of dielectric material of the first supporting structure **515** and the third supporting structure **550** may be monolithic, the one or more layers of dielectric material of the second supporting structure **535** and the fourth supporting structure **565** may be monolithic, and the dielectric material is polyimide, liquid crystal polymer, parylene, polyether ether ketone, or a combination thereof. The first supporting structure **515** further includes a proximal end **587** and a distal end **590**, the second supporting structure **535** further includes a proximal end **592** and a distal end **594**, the first electrode assembly **545** further includes a proximal end **595** and a distal end **596**, the second electrode assembly **560** further includes a proximal end **597** and a distal end **598**, the distal end **590** of the first supporting structure **515** is continuous with the proximal end **595** of the first electrode assembly **545**, and the distal end **594** of the second supporting structure **535** is continuous with the proximal end **597** of the second electrode assembly **560**. It should be understood that although only two subpanels are discussed with respect to lead assembly **500**, the use of any number and combination of subpanels is contemplated and applicable to all leads and devices that need electrodes/sensors that need to be attached to a neurostimulator.

[0061] FIG. 6 shows that a lead assembly may further include connector (e.g., connector **120** described with respect to FIG. 1). In some instances, a lead assembly **600** (e.g., lead assembly **400** described with respect to FIGS. 4A and 4B) may include a single connector body **605** formed on the proximal end **610** of a cable **615**. The single connector body **605** may include a first set of contacts or bond pads **620** in electrical connection with a first set of conductive traces **625** and a second set of contacts or bond pads **630** in electrical connection with a second set of conductive traces **635**. Alternatively (not shown), a lead assembly **600** (e.g., lead assembly **400** described with respect to FIGS. 4A and 4B) may include multiple connector bodies formed on the proximal end **610** of a cable **615**. Each connector bodies may include a set of contacts or bond pads in electrical connection with a set of conductive traces. Alternatively (not shown), a lead assembly **600** (e.g., lead assembly **400** described with respect to FIGS. 4A and 4B) may include multi-branched connector (see for example branched connector **645**) formed on the proximal end **610** of a cable **615**. Each branch or plug of the multi-branched connector may include a set of contacts or bond pads in electrical connection with a set of conductive traces.

[0062] In some instances, a lead assembly **640** (e.g., lead assembly **500** described with respect to FIGS. 5A, 5B, 5C, and 5D) may include multiple connector bodies **645** formed on the proximal ends **650** of multiple cables **655**. Each connector body **645** may include a main body **660** including

a supporting structure **665** and a set of conductive traces **670**, and a plurality of branches or plugs **675** extending from the main body **660**. Each plug of the plurality of plugs **675** includes: an end portion of the supporting structure **665** including the one or more layers of dielectric material and a subset of conductive traces **680** from the set of conductive traces **670**. Each trace from the subset of conductive traces **680** terminates at a contact or bond pad **685** exposed on a surface of the end portion of the supporting structure **665**. Alternatively (not shown), a lead assembly **640** (e.g., lead assembly **400** described with respect to FIGS. **4A** and **4B**) may include one or more connector bodies (for example see connector **605**) formed on the proximal ends **650** of cables **655**. Each connector bodies may include a set of contacts or bond pads in electrical connection with a set of conductive traces.

[0063] FIG. **6** further shows the supporting structures of the cable **615** may have one or more helical portions **690**. The supporting structures of the cable **615** may be arranged to run in parallel adjacent to one another within the cable **615**. In some instances, the cable **615** further includes a housing **695** encasing the helical portions **690** of the supporting structures. Moreover, the supporting structures of the cables **655** may have one or more helical portions **690**. In some instances, the cables **655** further include a housing **695** encasing the helical portions **690** of the supporting structures.

[0064] While the invention has been described in detail, modifications within the spirit and scope of the invention will be readily apparent to the skilled artisan. It should be understood that aspects of the invention and portions of various embodiments and various features recited above and/or in the appended claims may be combined or interchanged either in whole or in part. In the foregoing descriptions of the various embodiments, those embodiments which refer to another embodiment may be appropriately combined with other embodiments as will be appreciated by the skilled artisan. Furthermore, the skilled artisan will appreciate that the foregoing description is by way of example only, and is not intended to limit the invention.

Claims

1. A thin-film lead assembly comprising: a cable comprising: (i) a first supporting structure formed of one or more layers of dielectric material, (ii) a first set of conductive traces deposited on a surface of a portion of the first supporting structure, wherein the first set of conductive traces is deposited by one from among a sputter deposition, a chemical vapor deposition, an electroplating, and an electroless plating, (iii) a second supporting structure formed of one or more layers of dielectric material, and (iv) a second set of conductive traces deposited on a surface of a portion of the second supporting structure, wherein the second set of conductive traces is deposited by one from among the sputter deposition, the chemical vapor deposition, the electroplating, and the electroless plating; an electrode assembly comprising: (i) a third supporting structure formed of one or more layers of dielectric material, (ii) a first set of electrodes in electrical connection with the first set of conductive traces, and (iii) a second set of electrodes in electrical connection with the second set of conductive traces; and a connector comprising: (i) a fourth supporting structure formed on an end of the first supporting structure and an end of the second supporting structure, (ii) a first set of contacts or bond pads formed on at least a portion of the fourth supporting structure in electrical connection with the first set of conductive traces, and (iii) a second set of contacts or bond pads on at least another portion of the fourth supporting structure in electrical connection with the second set of conductive traces.
2. The thin-film lead assembly of claim 1, wherein the first set of electrodes and the second set of electrodes are arranged on the third supporting structure in a honeycomb pattern.
3. The thin-film lead assembly of claim 2, wherein the first set of electrodes and the second set of electrodes are arranged on the third supporting structure in a plurality of alternating rows comprising a first number of electrodes n and a second number of electrodes $n-1$.

4. The thin-film lead assembly of claim 3, wherein the electrode assembly further comprises a proximal end and a distal end, the first supporting structure and the second supporting structure connect with the electrode assembly at the proximal end, and a row of electrodes closest to the first supporting structure and the second supporting structure at the proximal end comprises a third number of electrodes $n-2$.
5. The thin-film lead assembly of claim 1, wherein the third supporting structure is thermoset into a cylindrical shape.
6. The thin-film lead assembly of claim 1, wherein the first set of conductive traces and the second set of conductive traces are comprised of one or more layers of conductive material, and the conductive material is copper (Cu), gold (Au), silver (Ag), gold/chromium (Au/Cr), platinum (Pt), platinum/iridium (Pt/Ir), titanium (Ti), gold/titanium (Au/Ti), or any alloy thereof.
7. The thin-film lead assembly of claim 1, wherein the one or more layers of dielectric material of the first supporting structure, the second supporting structure, and the third supporting structure are monolithic, and the dielectric material is polyimide, liquid crystal polymer, parylene, polyether ether ketone, or a combination thereof.
8. The thin-film lead assembly of claim 7, wherein the first supporting structure comprises a proximal end and a distal end, the second supporting structure comprises a proximal end and a distal end, the electrode assembly further comprises a proximal end and a distal end, the distal end of the first supporting structure and the distal end of the second supporting structure are continuous with the proximal end of the electrode assembly.
9. The thin-film lead assembly of claim 1, wherein the connector is a branched connector comprising: a first plug extending from the at least the portion of the fourth supporting structure, and a second plug extending from the at least the another portion of the fourth supporting structure.
10. The thin-film lead assembly of claim 1, wherein the first supporting structure comprise a first helical portion and the second supporting structure comprises a second helical portion, the first supporting structure and the second supporting structure run in parallel adjacent to one another within the cable.
11. The thin-film lead assembly of claim 10, wherein the cable further comprises a housing encasing the first helical portion of the first supporting structure and the second helical portion of the second supporting structure.
12. A thin-film lead assembly comprising: a first cable comprising: (i) a first supporting structure formed of one or more layers of dielectric material, and (ii) a first set of conductive traces formed on a portion of the first supporting structure; a second cable comprising: (i) a second supporting structure formed of one or more layers of the dielectric material, and (ii) a second set of conductive traces formed on a portion of the second supporting structure; a first electrode assembly comprising: (i) a third supporting structure formed of one or more layers of dielectric material, and (ii) a first set of electrodes in electrical connection with first set of conductive traces; and a second electrode assembly comprising: (i) a fourth supporting structure formed of one or more layers of dielectric material, and (ii) a second set of electrodes in electrical connection with the second set of conductive traces, wherein the first electrode assembly and the second electrode assembly are disposed on a backing formed of a polymer material.
13. The thin-film lead assembly of claim 12, wherein the third supporting structure and the fourth supporting structure are formed on the backing, the backing is formed of thermoplastic polyurethane, and the backing, the third supporting structure, and the fourth supporting structure are thermoset into a cylindrical shape.
14. The thin-film lead assembly of claim 13, wherein the one or more layers of dielectric material of the first supporting structure and the third supporting structure are monolithic, the one or more layers of dielectric material of the second supporting structure and the fourth supporting structure are monolithic, and the dielectric material is polyimide, liquid crystal polymer, parylene, polyether ether ketone, or a combination thereof.

15. The thin-film lead assembly of claim 13, wherein the first electrode assembly further comprises (iv) a first set of alignment holes formed in the third supporting structure, the second electrode assembly further comprises (iv) a second set of alignment holes formed in the fourth supporting structure, and the first set of alignment holes and the second set of alignment holes overlap with one another.

16. The thin-film lead assembly of claim 12, wherein the first supporting structure further comprises a proximal end and a distal end, the second supporting structure further comprises a proximal end and a distal end, the first electrode assembly further comprises a proximal end and a distal end, the second electrode assembly further comprises a proximal end and a distal end, the distal end of the first supporting structure is continuous with the proximal end of the first electrode assembly, and the distal end of the second supporting structure is continuous with the proximal end of the second electrode assembly.

17. A neuromodulation system comprising: a neurostimulator comprising an electronics module; a cable comprising: (i) a first supporting structure formed of one or more layers of dielectric material, (ii) a first set of conductive traces deposited on a surface of a portion of the first supporting structure, wherein the first set of conductive traces is deposited by one from among a sputter deposition, a chemical vapor deposition, an electroplating, and an electroless plating, (iii) a second supporting structure formed of one or more layers of dielectric material, and (iv) a second set of conductive traces deposited on a surface of a portion of the second supporting structure, wherein the second set of conductive traces is deposited by one from among the sputter deposition, the chemical vapor deposition, the electroplating, and the electroless plating; an electrode assembly comprising: (i) a third supporting structure formed of one or more layers of dielectric material, (ii) a first set of electrodes in electrical connection with the first set of conductive traces, and (iii) a second set of electrodes in electrical connection with the second set of conductive traces; a first connector formed on a proximal end of the first supporting structure, wherein the first connector comprises a first set of contacts or bond pads in electrical connection with the first set of conductive traces and the electronics module; and a second connector formed on a proximal end of the second supporting structure, wherein the second connector comprises a second set of contacts or bond pads in electrical connection with the second set of conductive traces and the electronics module.
